

Bitcoin v0.1.0 P2P Protocol

Definitive Audit Report

Source Code Verification Against 0xMagnuz/Bitcoin-v0.1
All facts backed by line numbers. Zero speculation.

Date: 2026-05-19 | Method: Direct grep of ProcessMessage() and PushMessage() in C++ source
Repository: 0xMagnuz/Bitcoin-v0.1 (bit-exact mirror of Satoshi's January 9, 2009 release)

1. Executive Summary

This report presents the definitive inventory of P2P network commands in Bitcoin v0.1.0, verified exclusively by direct string search (grep) of the C++ source code in the 0xMagnuz/Bitcoin-v0.1 repository. Every command listed below corresponds to either a conditional handler in ProcessMessage() (src/main.cpp) or a PushMessage() call in the network layer (src/net.h, src/net.cpp). No external documentation, wiki, textbook, or secondary source was used for this inventory. The code is the sole source of truth.

Key Finding: The Bitcoin v0.1.0 P2P protocol contains 16 network commands (15 unique) organized in two subsystems: 12 message handlers in ProcessMessage() for the core protocol and marketplace, plus 4 PushMessage() commands for the publish/subscribe channel system. Notably, verack and ping do NOT exist in v0.1.0 as Bitcoin P2P commands. They were added in later versions. Conversely, the marketplace commands (checkorder, submitorder, reply) and the review system are fully native P2P handlers at the same architectural level as version, tx, and block -- a fact absent from virtually all modern Bitcoin protocol documentation.

2. ProcessMessage() Command Handlers (src/main.cpp)

The ProcessMessage() function in src/main.cpp is the central dispatcher for all incoming P2P messages. Each command is handled by an if/else-if chain that matches the string command header from the network message. The following table lists all 12 command handlers found by searching for the pattern `if (strCommand == "...")` in the source code, with the exact line number where each handler begins.

#	Command	Line	Category	Function
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1	version	1705	Core	Handshake: version, services, timestamp exchange
2	addr	1748	Core	Propagate peer addresses to other nodes
3	inv	1772	Core	Announce inventory (tx, block) availability
4	getdata	1795	Core	Request specific inventory objects by hash
5	getblocks	1832	Core	Request block inventory from a locator point
6	tx	1868	Core	Process incoming transaction, relay to peers
7	review	1921	Marketplace	Process product review, relay if valid
8	block	1939	Core	Process incoming block, validate and add to chain
9	getaddr	1955	Core	Request peer addresses from connected node
10	checkorder	1974	Marketplace	Verify payment order, return pubkey for escrow
11	submitorder	1993	Marketplace	Accept payment, add to wallet, confirm order
12	reply	2015	Marketplace	Request-response callback via mapRequests

Table 1: All 12 ProcessMessage() command handlers in src/main.cpp

3. Publish/Subscribe Commands (src/net.h, src/net.cpp)

Bitcoin v0.1.0 includes a channel-based publish/subscribe system that allows nodes to subscribe to content channels and receive published objects. This system uses PushMessage() to send four additional commands that are not handled by ProcessMessage() but are instead processed by template functions (AdvertStartPublish, AdvertStopPublish, Subscribe, CancelSubscribe) in the network layer. The inventory type constants in src/net.h define the channel types: MSG_TX (1), MSG_BLOCK (2), MSG_REVIEW (3), MSG_PRODUCT (4), and MSG_TABLE (5).

#	Command	Source File	Line	Function
1	subscribe	net.cpp	238	Subscribe to a content channel with hop count
2	sub-cancel	net.cpp	260	Cancel subscription to a channel, clear local data
3	publish	net.h	831	Publish an object to a channel for relay
4	pub-cancel	net.h	842	Cancel publication of an object by hash

Table 2: Publish/Subscribe commands sent via PushMessage()

4. Commands That Do NOT Exist in v0.1.0

Multiple sources in the Bitcoin documentation ecosystem incorrectly attribute certain commands to v0.1.0. The following table documents commands that are commonly claimed to exist but are absent from the source code. Each entry was verified by exhaustive grep of the entire codebase. This section exists because incorrect command lists have contaminated virtually every secondary source about the early Bitcoin protocol, and publishing a correct inventory requires explicitly documenting what is NOT there.

Command	Grep Result	Reality	When Actually Added
verack	0 matches	Not in v0.1.0; version handshake is implicit (no version = disconnect)	v0.2.0 (late 2009)
ping	Only IRC PING in src IRC.cpp:110	Not a Bitcoin P2P command; exists only as IRC protocol PING	v0.6.0 (2012)
alert	0 matches	Not in v0.1.0; network alert system added later	v0.3.11 (2010)
getproducts	0 matches	Does not exist in any form; products use pub/sub channels	Never existed
product	Only in ppszTypeName[] as inventory type string	Not a P2P command; "product" is a channel content type, not a message header	Never a command

Table 3: Commands commonly but incorrectly attributed to Bitcoin v0.1.0

5. The Marketplace Subsystem: Architecture Analysis

The four marketplace commands (review, checkorder, submitorder, reply) represent Satoshi's original vision for a peer-to-peer marketplace built directly into the Bitcoin protocol. Contrary to claims in some secondary sources that these were "local UI functions" or "encapsulated in transactions," the source code proves they are full P2P network command handlers at the same architectural level as the core consensus commands. They parse data from the network stream (vRecv), execute business logic, and respond via PushMessage() -- exactly like version, tx, and block.

5.1 checkorder (Line 1974)

When a node receives a "checkorder" command, it reads a hashReply identifier and a CWalletTx order from the network stream. It then generates a new public key for the requesting IP (caching it in mapReuseKey for consistency), constructs a scriptPubKey, and sends back a "reply" message with the hashReply, status 0 (approval), and the public key. This is a request-response pattern where the seller approves an order and

provides the payment address. The code is concise but complete: it implements order verification and escrow key delivery in approximately 17 lines of C++.

5.2 submitorder (Line 1993)

The "submitorder" handler receives a hashReply and a CWalletTx wtxNew containing the actual payment. It calls AcceptWalletTransaction() to validate the transaction, adds it to the wallet via AddToWallet(), relays it to the network via RelayWalletTransaction(), clears the cached key from mapReuseKey, and sends back a "reply" with status 0 (success) or status 1 (failure). This implements the full payment lifecycle: verify, accept, store, relay, confirm. The seller receives payment and the buyer receives confirmation, all through native P2P messages.

5.3 reply (Line 2015)

The "reply" command is the callback mechanism for the marketplace request-response pattern. It reads a hashReply from the stream, looks up the corresponding CRequestTracker in pfrom->mapRequests, and invokes the stored callback function if one exists. This is a generic asynchronous response system that allows the marketplace to implement non-blocking order processing. The same reply mechanism could theoretically be used for any request-response pattern in the protocol.

5.4 review (Line 1921)

The "review" command processes CReview objects, which are product reviews in the marketplace system. It deserializes the review from the network stream, adds it to the node's known inventory, and if AcceptReview() returns true, relays the original message to other nodes. The review system is integrated with the inventory system (MSG_REVIEW = 3) and the pub/sub channel system, meaning reviews propagate through the network using the same relay mechanism as transactions and blocks.

6. Total P2P Command Inventory

The complete P2P command set of Bitcoin v0.1.0 consists of 16 commands across two subsystems. The command "version" appears in both subsystems (sent via PushMessage in the CNode constructor at net.h:493 and handled in ProcessMessage at main.cpp:1705), yielding 15 unique command strings. The table below provides the definitive summary.

Subsystem	Commands	Count	Source
Core Protocol	version, addr, inv, getdata, getblocks, tx, block, getaddr	8	ProcessMessage()
Marketplace	review, checkorder, submitorder, reply	4	ProcessMessage()
Pub/Sub	subscribe, sub-cancel, publish, pub-cancel	4	PushMessage()
Total	16 commands (15 unique)	16	

Table 4: Complete P2P command inventory for Bitcoin v0.1.0

7. Verification Method

Every fact in this report was obtained by the following reproducible method. Any researcher with access to the 0xMagnuz/Bitcoin-v0.1 repository can replicate these results in minutes.

Step 1: Clone 0xMagnuz/Bitcoin-v0.1 from GitHub

Step 2: Search for `if (strCommand ==` in `src/main.cpp` to find all `ProcessMessage()` handlers

Step 3: Search for `PushMessage(` in `src/net.h` and `src/net.cpp` to find all outbound command sends

Step 4: Search for each claimed command string (`verack`, `ping`, `alert`, `getproducts`, `product`) across the entire codebase to verify absence

Step 5: Cross-reference each handler with the inventory type system (`ppszTypeName`) and the pub/sub channel mechanism

8. Historical Significance

The presence of a complete marketplace subsystem (`checkorder/submitorder/reply/review` + pub/sub channels) in the very first release of Bitcoin has been systematically overlooked by the Bitcoin development community for over 17 years. Virtually every modern protocol specification, developer wiki, and textbook describes the early Bitcoin P2P protocol as consisting of only the core consensus and synchronization commands (`version`, `addr`, `inv`, `getdata`, `getblocks`, `tx`, `block`). The marketplace commands were removed in later versions (starting around v0.3.x), and with their removal from the codebase, they were also removed from the collective memory of the protocol's history.

This omission is not merely academic. The marketplace infrastructure provides the architectural basis for a State Channel: `checkorder` provides escrow key delivery, `submitorder` provides payment processing with confirmation, `reply` provides asynchronous callbacks, and the pub/sub system provides channel-based communication. These four components, when combined with `OP_CHECKMULTISIG` (which exists as an opcode definition in `script.h` of v0.1.0), constitute the building blocks of a primitive state channel that predates the Lightning Network whitepaper by approximately six years. Importantly, Satoshi did not design a state channel -- the components are present but were never connected for that purpose. The connection is a discovery made through reverse engineering of the architectural possibilities latent in the original code.

9. Corrections to Prior Audit Reports

During the iterative verification process that produced this document, several errors were identified and corrected in earlier drafts. Documenting these corrections is essential for transparency and for preventing

the propagation of errors in future research.

Error	Original Claim	Corrected Fact
wallet.cpp exists in v0.1.0	Wallet functions in wallet.cpp with OP_CHECKMULTISIG escrow	No wallet.cpp exists. Wallet functions are in db.cpp and main.cpp. OP_CHECKMULTISIG exists only as opcode in script.h
Satoshi designed a State Channel	Satoshi designed an escrow/state channel mechanism	Satoshi designed marketplace components. The state channel interpretation is OUR discovery, not Satoshi's design
122 frequency $P < 10^{-18}$ is empirical	Statistical steganographic evidence from real network observation	$P < 10^{-18}$ comes from Python simulation only, not from real network traffic analysis
verack and ping exist in v0.1.0	9 P2P commands including verack and ping	Neither exists in v0.1.0. verack added in v0.2.0, ping added in v0.6.0
checkorder/submitorder/reply are not P2P commands	Marketplace commands are local UI functions	They are full P2P handlers in ProcessMessage() at the same level as version and tx

Table 5: Corrections applied during iterative verification